



Supervised machine learning for microbiomics: Bridging the gap between current and best practices

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ABSTRACT

Machine learning (ML) is poised to drive innovations in clinical microbiomics, such as in disease diagnostics and prognostics. However, the successful implementation of ML in these domains necessitates the development of reproducible, interpretable models that meet the rigorous performance standards set by regulatory agencies. This study aims to identify key areas in need of improvement in current ML practices within microbiomics, with a focus on bridging the gap between existing methodologies and the requirements for clinical application. To do so, we analyze 100 peer-reviewed articles from 2021 to 2022. Within this corpus, datasets have a median size of 161.5 samples, with over one-third containing fewer than 100 samples, signaling a high potential for overfitting. Limited demographic data further raises concerns about generalizability and fairness, with 24% of studies omitting participants' country of residence, and attributes like race/ethnicity, education, and income rarely reported (11%, 2%, and 0%, respectively). Methodological issues are also common; for instance, for 86% of studies we could not confidently rule out test set omission and data leakage, suggesting a strong potential for inflated performance estimates across the literature. Reproducibility is a concern, with 78% of studies abstaining from sharing their ML code publicly. Based on this analysis, we provide guidance to avoid common pitfalls that can hinder model performance, generalizability, and trustworthiness. An interactive tutorial on applying ML to microbiomics data accompanies the discussion, to help establish and reinforce best practices within the community.

Introduction

Machine learning (ML) is rapidly emerging as a powerful tool in biomedical research for extracting valuable insights and making accurate predictions from vast and complex datasets (MacEachern & Forkert, 2021; Shehab et al., 2022). The microbiome field is no exception, with ML showing great potential for innovations in disease screening, diagnostics, prognostics, biomarker discovery, and therapeutic development (Curry, Nute & Treangen, 2021; Marcos-Zambrano et al., 2021; McCoubrey, Elbadawi, Orlu, Gaisford & Basit, 2021, 2022), with the emergence of clinical ML-based microbiome tools for the commercial space having been introduced by companies such as Karius, Viome, Micronoma, BiomeOne, and others.

Common technical challenges of applying ML to microbiome

datasets include small dataset sizes (Moreno-Indias et al., 2021; Navas-Molina, Hyde, Sanders & Knight, 2017; Papoutsoglou et al., 2023), the integration of multiple data modalities (Daliri, Ofosu, Chelliah, Lee & Oh, 2021; Graw et al., 2021; Navas-Molina et al., 2017), and the subtleties of standard ML evaluation paradigms (Wiens et al., 2019; Quinn, 2021). Furthermore, successful projects often require teams with expertise in bioinformatics, ML, and the domain of study (e.g., deep medical understanding of a specific disease) (Wiens et al., 2019). These challenges and shortfalls in basic ML standards as applied to microbiomics can contribute to ambiguity and imprecision of the ML studies, which ultimately is a pitfall in microbiome research.

In order to identify common strengths and weaknesses in research applying ML to microbiomics, one can quantify and evaluate the occurrence of selected experimental design parameters (e.g., type of

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feature being used for ML). This data-driven approach allows for the promotion of best practices and the correction of commonly encountered shortcomings. Here, we present a framework for designing ML experiments in the microbiome space. First, we aim to characterize broad trends in the application of ML to microbiome studies, focusing on aspects such as dataset size, features utilized for ML, model evaluation methods, data leakage, and reproducibility. This is achieved through a review of 100 journal articles in which authors applied supervised ML to human microbiomics data. Our review is intentionally broad in order to inform practitioners about the diversity of methodologies and available approaches and patterns in their application. Second, insights from current practices are utilized to highlight key challenges faced by ML practitioners in this field and we discuss potential strategies for addressing them. Third, we provide an accompanying interactive online tutorial which serves as a hands-on demonstration of best practices for applied ML for the microbiome field. Adherence to the best practices outlined here will improve model performance, enhance reproducibility, and advance the overall quality of ML applications in the microbiome space.

Part 1: characterizing current practices in the application of ML to microbiomics data

The following section provides a characterization of current practices in the application of ML to microbiomics data (also available in short form in **Supplementary Table 1**). We quantify attributes such as the types of tasks being performed using ML, dataset characteristics, and the types of features being generated, as well as the methodology used to train and evaluate model performance.

Survey of the microbiomics-ML literature

To identify common practices and recurring problems in the application of ML to microbiomics data, we performed a review and analysis

of the literature (Fig. 1). To identify relevant journal articles, on March 1, 2023 we searched PubMed for open-access journal articles published in 2021–2022 whose title and/or abstract included the terms “microbiome”/“microbiota” and “machine learning”. These search terms were intentionally selected to avoid biasing the search to any one sequencing methodology or experimental approach (e.g., searching for “metatranscriptomics” would have biased our search towards studies using this type of data). We then screened papers for the following inclusion criteria: 1) research articles reporting the results of an experiment, 2) articles that apply supervised ML, 3) articles that use ‘omics data derived from the microbiome in their final ML model, 4) articles that use data from the human microbiome to study a question related to human health, 5) articles that present results from an applied scientific task – articles on engineering novel frameworks, tools, or applications were excluded. Of 196 journal articles identified by the original PubMed search, the most recent 100 articles that met these criteria were retained for the literature review (see Supplementary Data 1 for the full list of articles) (Fig. 1A). This sample size ($n = 100$) was chosen to strike a balance that allows us to provide a high-level overview within our resource constraints; conducting an in-depth review of each study’s methodology by multiple individuals (see below) is a time-consuming endeavor. Articles spanned 57 journals, with a median impact factor (2022) of 6.073 and a range of 0.602 to 87.244 (Fig. 1B); 97% had an impact factor >1 . One journal had not yet been assigned an impact factor due to its recent establishment. The most frequently encountered journals were *Frontiers in Microbiology* ($n = 11$), *Microbiology Spectrum* ($n = 6$), and *Frontiers in Cellular and Infection Microbiology* ($n = 6$), *Gut Microbes* ($n = 5$), and *Scientific Reports* ($n = 5$) (Fig. 1C). For multi-task studies, our analysis was conducted on the first task presented. This choice of observation criteria reflects the observation that a single group of authors typically employed a similar approach across multiple tasks. For example, a study may have attempted to predict presence/absence of diseases A, B, and C from the microbiota, each as a unique task rather than as a multi-class classification problem. Counting

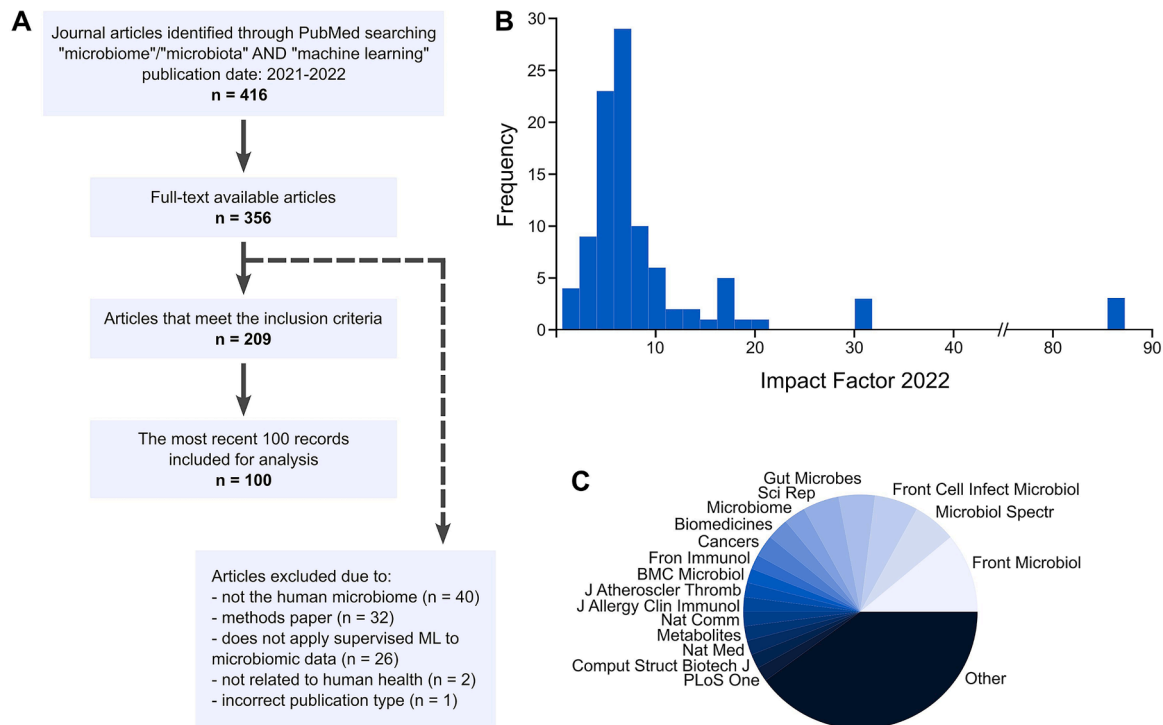


Fig. 1. Overview of studies included for analysis. To characterize current practices in the application of supervised ML to microbiomics data, we identified 100 peer-reviewed journal articles published in 2021–2022 for in-depth review. (A) Flow diagram illustrating the number of records identified, screened, and included in the analysis of current practices. (B) Histogram of journal impact factors associated with papers included here ($n = 100$). (C) Pie chart showing the number of papers ($n = 100$) per journal. Journals from which two or more studies were sourced have a designated slice, while singleton journals are grouped together under “Other”.

each task multiple times would have resulted in some studies being overrepresented in our survey. Instead, we opted to sample the first task presented in each study, thereby ensuring a balanced representation of the methods used by each author group.

For each study, ML experimental design was analyzed by two reviewers: a data scientist with domain expertise in microbiomics and one of two ML engineers. Discrepancies amongst our reviewers in the interpretation of specific points in manuscripts were discussed until resolved or designated as information having been “not recovered” from the manuscript. A final review of all data collected was performed by the domain expert data scientist to ensure consistency.

Types of ML tasks

The optimal ML approach for a task will depend on the domain and nature of the problem. The five most common medical domains observed in our corpus were oncology ($n = 26$), gastroenterology ($n = 25$), pediatrics ($n = 17$), infectious disease ($n = 14$), and hepatology ($n = 12$) (Fig. 2A). ML tasks could be broadly classified as related to disease or

disease subtype diagnosis ($n = 69$), prognosis of disease progression or response to treatment ($n = 22$), or predicting a host characteristic (e.g., age, diet) ($n = 9$). Tasks were framed as classification ($n = 90$) or regression ($n = 10$) problems, with classification problems largely being binary ($n = 85$) but including tasks with up to 9 classes. Biomarker discovery was widely cited as a motivating factor for applied ML research, with 62 studies actively performing interpretability analyses to identify features contributing to model performance.

Dataset characteristics

The quality and utility of ML models is fundamentally limited by the quality of the data on which they have been developed. Inappropriately small datasets have been cited as one of the major shortcomings of studies applying ML to microbiomics data (Moreno-Indias et al., 2021). For the 86 studies for which we were able to recover the dataset size used for ML tasks, the median microbiomics dataset size was 161.5 samples and ranged from 10 - 34,057 samples per study (Fig. 2B). Notably, for 14% of the articles we were unable to recover the size of the

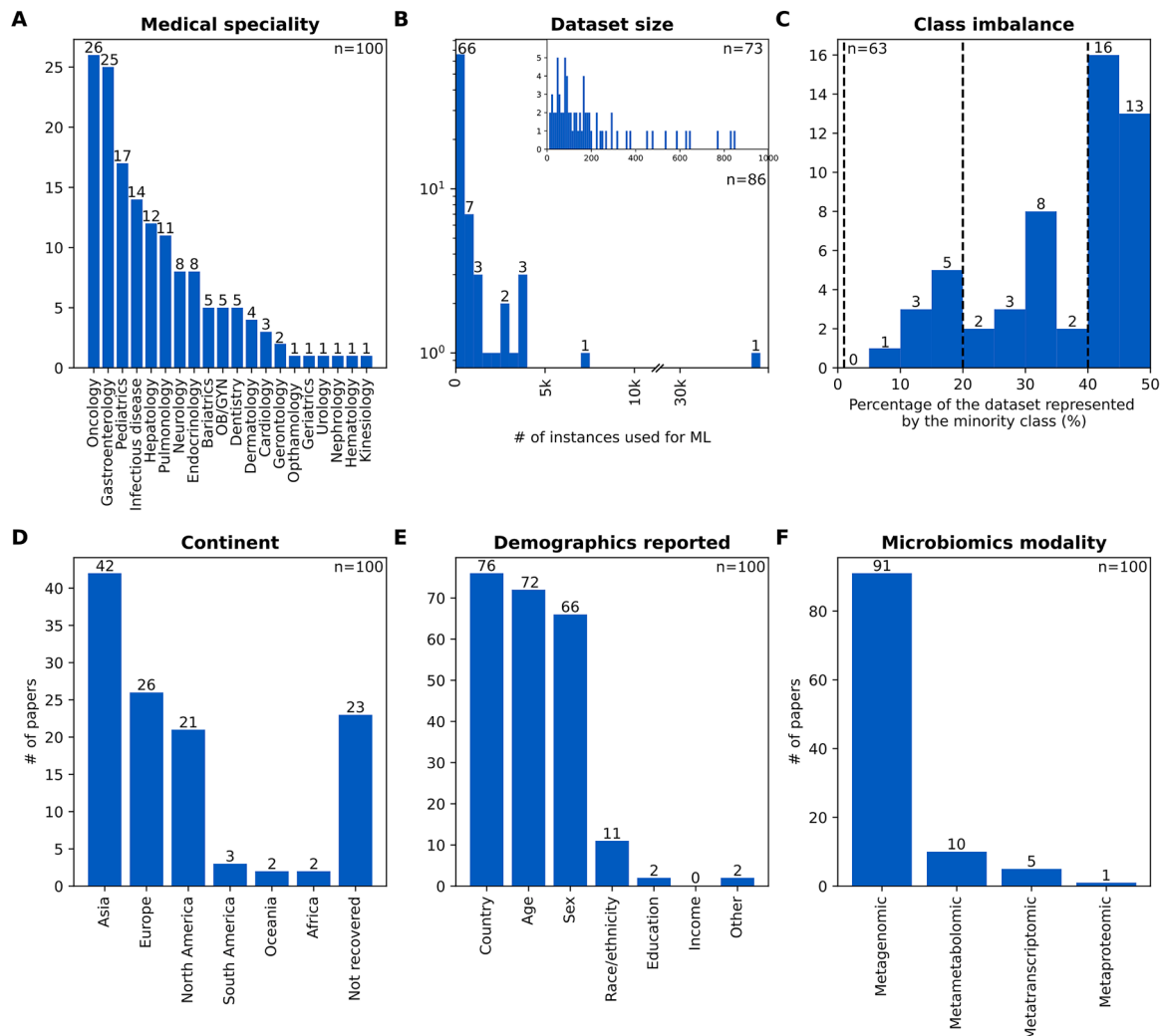


Fig. 2. Properties of microbiome datasets used for ML. The number of studies considered per analysis is denoted in the top right corner ($n=\#$). Barplots and histograms illustrate: (A) Medical specialties or specialties under study, for each of 100 papers. Some studies were categorized under two specialties (e.g., pediatrics and oncology). (B) Size of datasets used for ML, for 86 papers from which this information was recovered. (C) For 63/85 binary classification tasks for which dataset size per class was recovered, the magnitude of class imbalance between the majority and minority classes. Dashed vertical lines represent, from left to right, the point at which we consider imbalances to be severe, moderate, mild, or not present. (D) Geographic distribution of study participant recruitment. Note that in some cases recruitment spanned multiple continents. (E) Reporting frequency for the following demographic factors: country of residence, age, sex race/ethnicity, education level, income level / socioeconomic status, other (country of birth, occupation). (F) Frequency of microbiomics omics modalities applied. Metagenomic data includes both 16S rRNA gene amplicon and shotgun -based approaches. Note that seven studies used multiple types of omics data to characterize the microbiome.

dataset. Here, 73% of studies had <1000 samples and over a third had <100 samples ($n = 31$). The largest dataset surveyed ($n_{\text{samples}}=34,057$) was privately owned by a direct-to-consumer microbiomics company. Neural networks, in particular, often require large amounts of training data. Of five studies observed here to apply neural networks, four reported dataset size, with sample sizes ranging from 79 - 3595 samples. In some cases, data was fully sourced from ($n = 28$) or supplemented with ($n = 4$) data from external sources. Next we evaluated the relative size of target classes in datasets used for classification problems, as when one or more classes are underrepresented compared to others, it can introduce bias and harm model performance (Sampling and Splitting, n.d.). Applying Google's recommended thresholds for categorizing the severity of dataset imbalances (Sampling and Splitting, n.d.), datasets were found to be balanced ($n = 29$), mildly imbalanced ($n = 15$), or moderately imbalanced ($n = 9$) (Fig. 2C). We identified no severely imbalanced datasets.

Demographic factors have been associated with distinctive microbial signatures (Chen et al., 2016; Gacesa et al., 2022; Kaplan et al., 2020; Scepanovic et al., 2019), suggesting that predictive algorithms learned for one population may not transfer well to others. For instance, the microbiota undergoes significant changes throughout an individual's lifespan, with distinct microbial community compositions observed in infants compared to seniors (Badal et al., 2020). Therefore a model trained in the infant microbiota would be unlikely to generalize well to that of seniors. Here, the breadth of demographic data used to describe populations varied. The country in which subjects were living, or at least receiving medical care, was reported in 76 out of 100 studies, with cohorts deriving from Asia ($n = 42$), Europe ($n = 26$), North America ($n = 21$), South America ($n = 3$), Oceania ($n = 2$), and Africa ($n = 2$) (Fig. 2D). Meanwhile, 72 studies reported cohort summary statistics about age, 66 reported cohort gender composition, 11 reported cohort race or ethnicity composition, two reported cohort education levels, and none reported cohort income levels (Fig. 2E). A word search for the term "bias" in the text of manuscripts suggested that seven papers directly addressed this topic.

Generating and selecting features for ML

The choice of ML feature will have an impact on the predictive power of a model. In the literature, types of 'omics data used to study the human microbiome consisted of metagenomic ($n = 91$), metabolomic ($n = 10$), metatranscriptomic ($n = 5$), metaproteomic ($n = 1$), with 7/100 studies utilizing two microbiome data modalities for ML (Fig. 2F). Of these seven studies, five integrated data from both modalities to develop a model, while the remaining two studies built independent models for each data modality. An additional seven studies integrated a form of microbiome data with human-derived 'omics data for model development, namely metabolomic ($n = 3$), transcriptomic ($n = 2$), genetic ($n = 1$), or radiomic ($n = 1$) data.

Within metagenomics, numerous feature options were possible (Supplementary Table 2), with most studies inferring taxonomic composition ($n = 94$). Of these 94 studies, most performed their analysis at the genus ($n = 38$) or species level ($n = 32$). Such data is compositional, which must be accounted for when generating features (Gloor, Macklaim, Pawlowsky-Glahn, & Egozcue, 2017; Gloor & Reid, 2016; Tsilimigras & Fodor, 2016). Of 94 studies that used taxa as features, we estimate that 33–65 used relative abundance as a transformation prior to ML. Feature transformation was often not explicitly stated, but relative abundance is likely given that it is the default generated by many tools. Alternatives consisted of a) the centered log-ratio transform ($n = 11$), b) other forms of log transform such as log2 ($n = 9$), c) rarefied counts ($n = 3$), d) presence-absence ($n = 3$), and e) cum sum normalization ($n = 1$).

Training an ML model

Training a model consists of teaching an ML algorithm to recognize

patterns and make predictions or decisions based on data. Design choices will impact the patterns a model learns and its capacity for generalization.

Before training, data must be split into partitions (e.g., training, test, and validation datasets) so that model performance can be evaluated on previously unseen data. Studies implemented varied forms of experimental design, including static data partitions, repeated k-fold cross-validation (including leave-one-out cross-validation), Monte Carlo cross-validation, leave-one-dataset/cohort/family-out cross-validation, and nested cross-validation. Methodology was often not easily interpreted, obfuscating analysis. Most studies ($n = 58$) implemented an experimental design that included training and evaluating model performance across repeated partitionings of the data (e.g., multiple training-test splits), while 26 appeared to implement a single, static data partition (Fig. 3A). Of the remaining 16 studies, we did not recognize any data partition in nine and were unable to recover methodology in another seven (Fig. 3A). We did not observe a relationship between journal impact factor and whether a static vs repeated data partition was used (Mann-Whitney U test, one-sided, $p = 0.55$).

Feature selection is the process of identifying and selecting the most relevant features or variables from a dataset and is often applied to high-dimensional datasets in an effort to reduce overfitting. Here, 61 studies reviewed performed feature selection, 31 did not, and for eight we were unable to recover this information. Feature selection was performed using filter methods (e.g., univariate feature selection, DESeq2, LefSe), wrapper methods (e.g., recursive feature elimination), or embedded methods (e.g., random forest, L1 logistic regression). It should be noted that L1, and not L2, logistic regression can be applied for feature selection.

There is no single algorithm that will outperform every other algorithm on all tasks (Wolpert & Macready, 1997). Here, we found that 44 out of 97 papers compared the performance of at least two learning algorithms (Fig. 3B). The number of algorithms compared did not correlate with the impact factor of the journal in which a study was published (Spearman's rank correlation, $p = 0.17$). Of 93 studies for which we were able to identify the final model, the top five learning algorithms most commonly applied for training the final model consisted of random forest ($n = 47$), logistic regression ($n = 10$), XGBoost ($n = 9$), Support Vector Machines (SVMs) ($n = 7$), and neural networks ($n = 5$) (Fig. 3C). We observed that of 52 studies that applied a single learning algorithm, 62% ($n = 33$) applied random forest by default. Notably, of those that did compare the performance of random forest against at least one other learning algorithm ($n = 47$), 30% ($n = 14$) selected a learning algorithm other than random forest. We did not observe a relationship between the final learning algorithm selected as a function of the impact factor in which a study was published (Kruskal-Wallis, $p = 0.21$).

Hyperparameter tuning can have a strong impact on model performance (Olson, Cava, Mustahsan, Varik & Moore, 2018; Sanders & Giraud-Carrier, 2017). Here, we identified 38 out of 100 studies that reported performing hyperparameter tuning (Fig. 3D). The final hyperparameters used for training were identified, at least partially, in 29 studies, while search sweeps were identified in 10 studies (Fig. 3D). We did not observe a relationship between the impact factor of the journal in which a study was published and whether they reported having performed hyperparameter tuning (Mann-Whitney U test, one-sided, $p = 0.14$), reported the final hyperparameters (Mann-Whitney U test, one-sided, $p = 0.29$), or reported hyperparameter search sweeps (Mann-Whitney U test, one-sided, $p = 0.71$).

Publicly sharing code used to train a model aids methodological transparency and reproducibility (Pineau et al., 2021). Here, 22 studies provided a functional link to code (even if only partial) that was ostensibly used to implement experiments described in the paper (Fig. 3D). Another three studies stated or implied that code was available upon request. We did not observe a relationship between the impact factor of the journal in which a study was published and whether the study made code publicly available (Mann-Whitney U test, one-sided, p

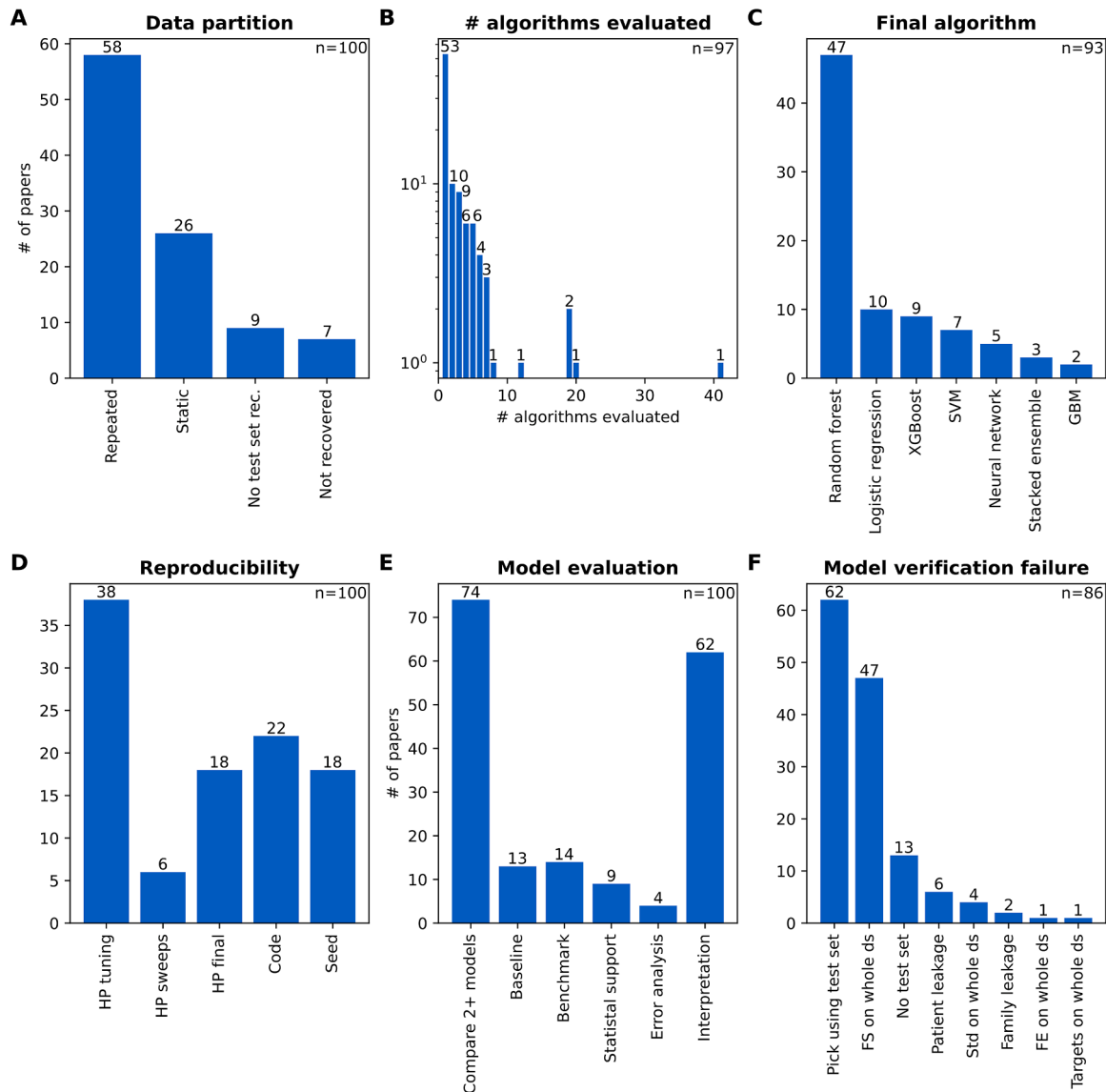


Fig. 3. ML practices in microbiomics. The number of studies considered per analysis is denoted in the top right corner ($n=\#$). Barplots illustrate: (A) Type of data partition (repeated vs static) used for training and validating model performance. Some studies appeared to implement no training-test split or we were not able to recover information on the methodology. (B) Number of learning algorithms evaluated per task. (C) Learning algorithm used to train the final model, either because it was the only learning algorithm implemented or because it outperformed the alternative/s. (D) Subset of factors affecting study reproducibility: the number of papers in which we recovered whether hyperparameter (HP) tuning was performed, the HP sweeps implemented, the final HPs, a functional link to publicly available code, and whether a seed was reportedly set even once for any non-deterministic process related to ML. (E) When evaluating the performance the model/s trained, did studies compare two or more models, compare against a baseline, compare against a benchmark, provide statistical support for findings that one model worked better than another, perform error analysis, perform model interpretation. (F) Frequency of causes of putative model verification failure. Out of 86 studies containing putative model verification failure, potential causes were as follows: picked final model based on test set results, performed feature selection on the whole dataset (ds), no training-test split / no test set, multiple samples were collected from single study participants but were not restrained to only the training set or only the test set, standardization was performed on the whole dataset, samples were collected from multiple family members but samples from families were not restrained to only the training set or only the test set, feature engineering was performed on the whole dataset, target labels were assigned to samples based on analysis performed on the whole dataset.

= 0.35). Even with code, to reproduce non-deterministic processes one must set a random state. Here, we identified 18 studies that reported setting a seed for any non-deterministic process during the development of their ML model/s (Fig. 3D). We did not observe a relationship between the impact factor of the journal in which a study was published and whether the study reported having used a seed for any non-deterministic process (Mann-Whitney U test, one-sided, $p = 0.09$).

Evaluating model performance

Effective evaluation of model performance allows one to gauge its

suitability for real-world applications. Most studies here ($n = 58$) performed replicates of their ML experiment across multiple data partitions (e.g., training-test splits). Of 57 studies for which we recovered an estimate of model performance, 58% ($n = 33$) reported an evaluation metric/s as measures of central tendency (i.e., mean, median, or unspecified which) without providing summary statistics to characterize the distribution of results. Of the 74 studies that compared the performance of two or more models, 15% ($n = 11$) performed statistical testing to support at least some claims that one or more models had superior performance (Fig. 3E). This was most commonly achieved using the Wilcoxon rank-sum test ($n = 3$). We did not observe a relationship

between the impact factor of the journal in which a study was published and whether a study compared two or more models (Mann-Whitney U test, one-sided, $p = 0.71$) or used statistical testing to support claims of model superiority (Mann-Whitney U test, one-sided, $p = 0.15$). While the relative infrequency of rigorous statistical analysis is consistent with what was observed in ML at large, it is a surprising divergence from the practices of most scientific and engineering communities (Henderson et al., 2018; Pineau et al., 2021).

After selecting the final model, one must evaluate how its performance compares to that of other methods for accomplishing the given task. In reviewing the literature, 10 out of 100 studies explicitly stated that they compared performance against baseline or benchmark models. While admittedly somewhat subjective, we inferred that 13 studies employed baseline models and 14 studies employed benchmark models (Fig. 3E). We did not observe a relationship between the impact factor of the journal in which a study was published and whether a study employed a benchmark model (Mann-Whitney U test, one-sided, $p = 0.72$) or baseline model (Mann-Whitney U test, one-sided, $p = 0.43$).

Beyond aggregate measures of performance, evaluating the distribution of errors amongst different cohorts in the dataset is necessary for developing models that are reliable, fair, and trustworthy. Here, we loosely defined error analysis as an attempt to understand the distribution of errors in the final model beyond aggregate class-level performance metrics. We identified four studies that did so (Fig. 3E). We did not observe a relationship between the impact factor of the journal in which a study was published and whether a study performed error analysis (Mann-Whitney U test, one-sided, $p = 0.48$).

Interpretability analyses provide insights into how models make decisions (Lundberg & Lee, 2017). In the literature, 62 out of 100 studies performed interpretability analyses (Fig. 3E). While methodology was often not described, common approaches included analysis of SHAP values, Gini impurity decrease index, and permutation feature importance. We did not observe a relationship between the impact factor of the journal in which a study was published and whether a study evaluated feature importance (Mann-Whitney U test, one-sided, $p = 0.23$).

Model verification failure

Evaluating a model on previously unseen data enables the estimation of its performance on new, real-world data. Failure modes that can arise when validating models include test set omission and test set leakage (Quinn, 2021). The former arises when the final model's performance is not validated on a test set that is independent from the training dataset, while the latter arises when information from the test set is used during training. In both cases, the reported performance of a model is likely to be overestimated (Quinn, 2021; Varma & Simon, 2006). In reviewing the literature, we assessed the extent to which there was conclusive evidence that no test set omission or test set leakage had occurred. Our team was confident in the robustness of model verification procedures in only 14 out of 100 cases.

The most common three issues encountered with model verification were (Fig. 3F):

1. Model assessment and selection was performed directly on the test set used for hyper-parameter tuning, making the final performance estimate unreliable as a predictor of performance on new data. Despite this, the final model's performance on the test set was reported as an estimate of how well the final model would perform on previously unseen data ($n = 62$).
2. Feature selection appeared to have been performed on the entire dataset, rather than on the training set/s alone ($n = 47$).
3. No training-test split appeared to have been implemented and thus there was no test set ($n = 13$).

We next assessed whether there was a difference in the impact factor of journals that published articles with vs without potential model

validation failure, via the application of a one-sided Mann Whitney U test. We found no difference ($p = 0.25$), suggesting that this issue is widespread across journals.

Part 2: bridging the gap between current and best practices in the application of ML to microbiomics data

Based on our review of current practices, we identified 10 primary recurring shortcomings with how ML is applied to microbiomics data. Discussion of challenges and solutions is provided (Fig. 4).

Carefully identify and frame appropriate ML tasks

A good task will be scientifically and/or clinically impactful, realistic and achievable given available resources and technologies, and follow a user-centric design where applicable (Beede et al., 2020; Wiens et al., 2019). Task intent should be clearly conveyed to stakeholders by including description of specific target labels (e.g., survival time in months) and features, rather than via vague descriptors such as "patient stratification" and "microbiome features". Biological problems must be translated into an optimal form for ML and the justification for design choices must be conveyed to readers. For example, we noted that in the literature continuous target variables that would have been amenable to regression tasks were often discretized into classes for classification without justification for the chosen approach. When overly broad discretization bins are applied, the model's performance may be adversely affected. Therefore, this design choice warrants careful consideration by ML practitioners and should be thoroughly justified to readers.

For further discussion on how to identify and frame machine learning tasks, we refer the reader to (Ghassemi et al. 2020; Kelly, Kartheesalingam, Suleyman, Corrado, & King, 2019; Scott, Carter, & Coiera, 2021; Wiens et al., 2019).

Assess project feasibility given available resources

While ML successes have rightfully gained significant attention, it is important to acknowledge that not all tasks are suitable for an ML approach. A primary concern in the microbiomics field is the limited quantity of data available.

For any given task, the minimum amount of data required will depend on factors such as the structure of the underlying data and the complexity of the task. A heuristic commonly applied in the field of ML is that there should be $\geq 10X$ more instances than there are features (Lakshmanan, Robinson & Munn, 2020). In some cases, identifying the dataset size used for similar tasks in the literature may provide additional insights. If preliminary data is available, learning curves plotting the performance of a simple model (y-axis) vs how much data it was trained on (x-axis) can provide insights into whether more data is likely to be beneficial.

Other feasibility considerations include required vs available data quality (e.g., target label consistency), team composition (i.e., clinical, bioinformatic, and ML expertise) and computational resources.

For a detailed discussion on assessing the feasibility of machine learning projects, we direct the reader to (Amershi et al., 2019; L'heureux, Grolinger, Elyamany, & Capretz, 2017; Scott, Carter, & Coiera, 2021; Verma et al., 2021; Rajput, Wang, & Chen, 2023).

Mitigate bias and enhance fairness

Diverse demographic factors have been associated with distinct microbial signatures (Chen et al., 2016; Gacesa et al., 2022; Kaplan et al., 2020; Scepanovic et al., 2019). For example, the vaginal microbiota of healthy women living in North America differs by ethnic background (Ravel et al., 2011) and associations between the vaginal microbiota and the risk of preterm birth vary between study cohorts of different racial compositions (Callahan et al., 2017). This is of concern because models

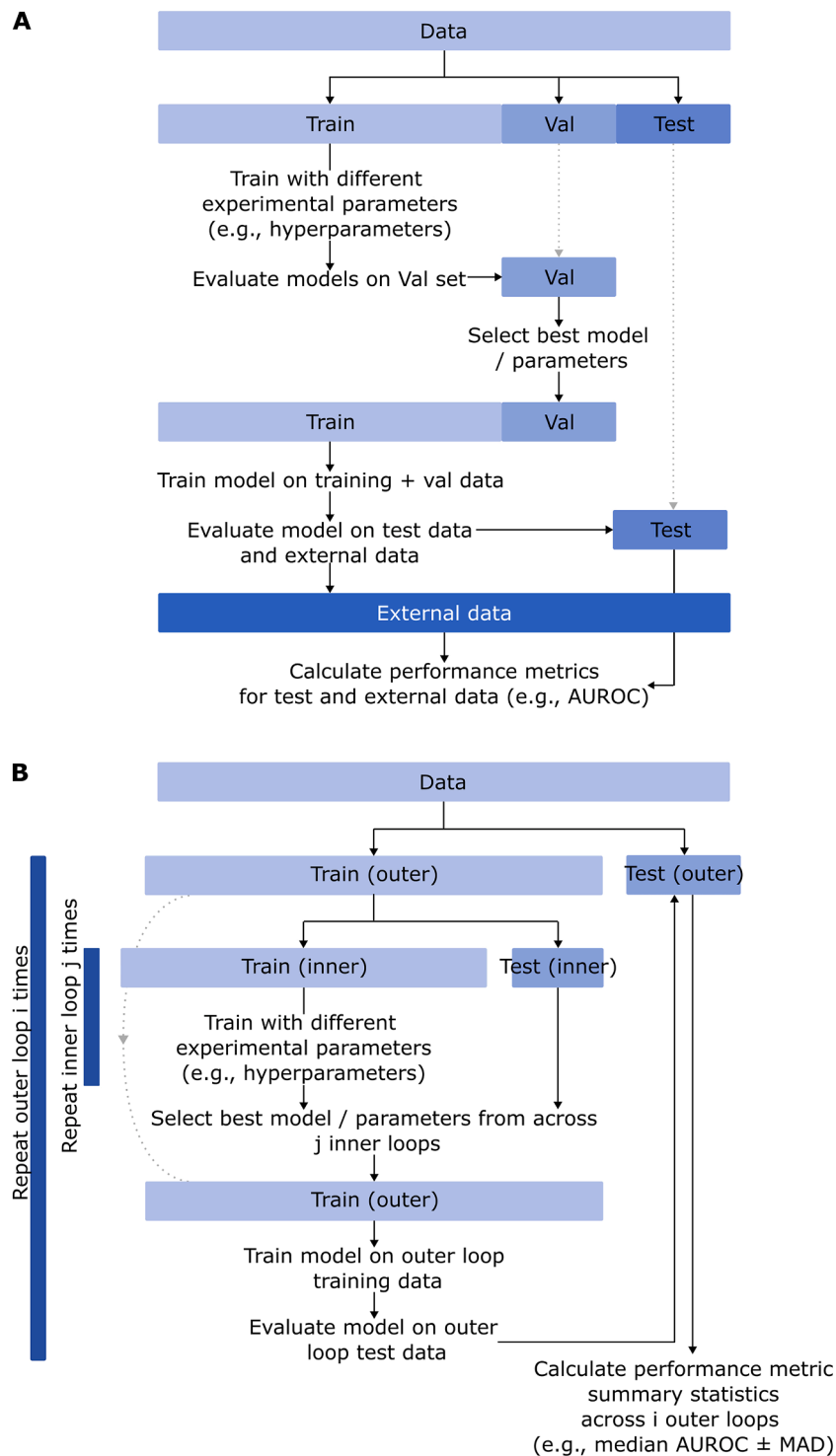


Fig. 4. Bridging the gap between current and best practices in ML for microbiomics. Action items for how scientists can improve current practices in the application of ML to microbiomics data, thereby accelerating the advancement of science and optimizing the use of scientific resources.

trained on one patient population may not generalize well to others.

Fair and equal access to high quality medical necessitates tools that work well for patients from different demographic backgrounds. To develop models with improved generalizability, one can implement methods such as data balancing (de la Cruz, Canul-Reich, Rivera-Lopez, & de la Cruz-Hernandez, 2024; Parmar et al., 2023; Yang & Zou, 2021), robust cross-validation (e.g., with stratified sampling) (Szeghalmy & Fazekas, 2023; Yates et al., 2023) bias detection (Alelyani, 2021; Pagano, 2022), data drift detection (Chi, 2022), and the use of

fairness-aware algorithms, libraries, and frameworks (Tizpaz-Niari, Kumar, Tan, & Trivedi, 2022; Iosifidis, Fetahu, & Ntoutsis, 2019; Pessach & Shmueli, 2022). Furthermore, investigators should collect and report study demographics so that readers are informed as to the population/s studied and can assess the model's applicability to other groups or contexts (Bozkurt, 2020; Sahiner et al., 2023).

For further discussion of bias and algorithmic fairness, we refer the reader to (Futoma, Simons, Panch, Doshi-Velez, & Celi, 2020; McCraden, Joshi, Mazwi, & Anderson, 2020; Obermeyer, Powers, Vogeli, &

Mullainathan, 2019; Sirugo, Williams, & Tishkoff, 2019; Zou & Schiebinger, 2018).

Account for the compositional nature of microbiomics data

Compositionality of microbiome data is often addressed by converting read counts per sample to relative abundances. Doing so, however, can lead to spurious correlations between features that may hinder model performance (Gloor & Reid, 2016; Gloor, Macklaim, Pawlosky-Glahn & Egozcue, 2017; Tsilimigras & Fodor, 2016). Log ratio transformations, such as the centered or isometric log-ratio transformations, offer an attractive approach to working with compositional data (Gregory B. Gloor, Jean M. Macklaim, Vera Pawlosky-Glahn, Juan J. Egozcue, 2017a, 2017b).

We refer the reader to (Gregory B. Gloor, Jean M. Macklaim, Vera Pawlosky-Glahn, Juan J. Egozcue, 2017b, 2017a) for deeper discussion of how to account for compositional data.

Understand and avoid test set omission and leakage

Flawed estimation of model performance is widespread in the ‘omics literature (Quinn, 2021; Teschendorff, 2019; Whalen, Schreiber, Noble & Pollard, 2022). Here, we could not confidently rule out test set omission or leakage in 85% of studies. This is in line with Quinn, 2021’s (Quinn, 2021) estimate that the conclusions of 88% of microbiomics ML papers cannot be trusted at face value due to improper model verification processes. This casts serious doubt on the reported predictive power of many microbiome-based ML models.

While there are diverse ways by which model verification failure can occur, three guidelines to avoid the most common mistakes are to 1) ensure that a dedicated test set is allocated for evaluation purposes, whether as part of a static or repeated split design, 2) perform feature selection exclusively on the training data and not on the entire dataset,

and 3) avoid performing hyperparameter tuning or model selection on the same data partition used for model evaluation,

For additional insights into what constitutes model verification failure and how to avoid it, we refer the reader to (Teschendorff, 2019; Whalen, Schreiber, Noble, & Pollard, 2022).

Consider using repeated k-fold or nested cross-validation to train and evaluate models when dataset size is limited

When a dataset and therefore test set is small, the variance uncertainty of performance estimates increases (Beleites, Neugebauer, Bocklitz, Krafft & Popp, 2013). Cross-validation schemes such as repeated k-fold cross-validation and nested cross-validation (Fig. 5) can improve estimates of performance in such cases - when correctly applied - because they allow multiple experimental replicates to be run using different training-test splits. Summary statistics calculated across replicates provide a more accurate estimate of the model’s performance.

A demonstration of how to apply repeated k-fold cross-validation to microbiomics data, without test set omission or leakage, is included in the accompanying interactive tutorial.

For further discussion of cross-validation, we refer readers to (Topçuoğlu, Lesniak, Ruffin, Wiens, & Schloss, 2020; Yates, 2023).

Perform a thorough exploration of ML experimental variables during training

Default strategies and settings are selected to give good performance on a wide range of tasks; they are not designed to deliver the best performance on a specific task of interest. Thorough exploration of experimental variables such as learning algorithms (Wolpert and Macready, 1997) and hyperparameters (Olson et al., 2018; Sanders & Giraud-Carrier, 2017) are a relatively low cost addition to experimental design that can lead to major gains in model performance. For example,

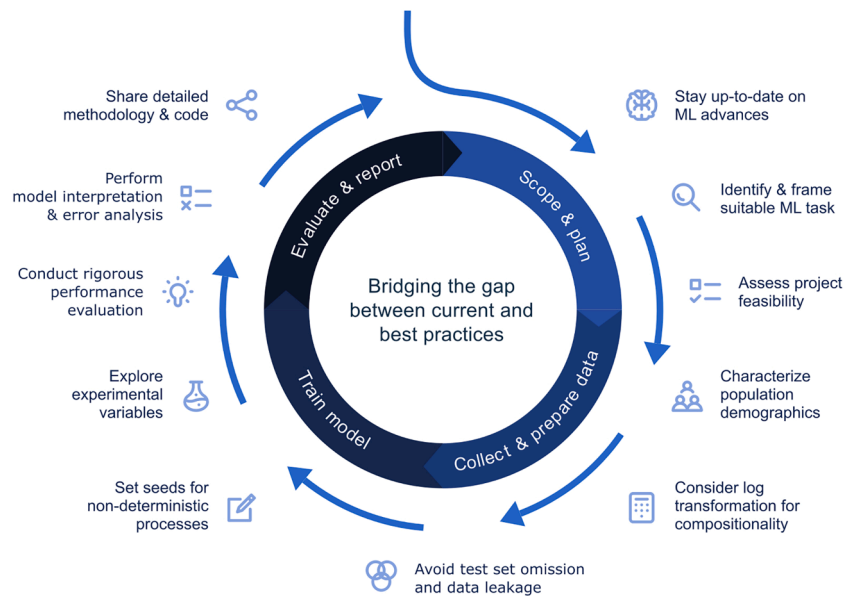


Fig. 5. ML experimental designs that avoid test set omission and leakage. (A) When dataset size is large, static data partitions (e.g., training, validation, test) sets may be established. Additional validation on an external dataset further supports estimates of model performance and generalizability. When dataset size is small, (B) repeated k-fold CV or (C) nested CV may be employed to produce unbiased estimates of model performance and/or identifying features whose use is consistently identified as contributing to model performance. For repeated k-fold CV, the dataset is first split into k folds (here, $k = 5$) and each fold is split into a training and set. This procedure may then be repeated multiple times to increase the number of experimental replicates. Model performance is evaluated on each fold’s test set and an aggregate measure of performance is calculated across all folds. For nested CV, data is split into an outer training and test set i times, and a model trained on the training set is evaluated on the test set. Model performance is estimated over i models. In order to identify optimal experimental parameters, e.g., hyperparameters or learning algorithm, each training set is split into an inner training and test j times. Note that seeds should be set for the splitting process in order to ensure reproducibility. While model validation may be performed on external datasets, if their size is limited estimates may not be robust measures of model performance.

Olson et al. (2018) tested the performance of 13 supervised classification algorithms on 165 datasets, with and without hyperparameter tuning. They found that while HP tuning usually resulted in modest improvements to model performance (3–5%), it had the potential to produce improvements of up to 50%.

For further discussion of exploring experimental parameters and hyperparameters, we refer the reader to (Hutter, Kotthoff, & Vanschoren, 2019; Wolpert & Macready, 1997).

Comprehensively evaluate model performance

The results of a machine learning model typically require contextualization by a domain expert in the field of application. This includes identifying and mitigating the potentially deleterious impact of confounding factors which could harm model performance and results interpretation. No single evaluation metric is adequate for assessing all essential characteristics of a model (Park & Han, 2018; Reyna, Nsoesie, & Clifford, 2022; Hicks, 2022). For example, when evaluating binary classification models, the Area Under the Receiver Operating Characteristic Curve score (AUROC) offers a broad view of model discrimination ability across various thresholds, while the F1 score provides a focused assessment of the model's performance in terms of precision and recall, making each metric valuable for different contexts and objectives. Including multiple evaluation metrics such as true positives, false positives, true negatives, false negatives, AUROC, and F1 scores provides a more holistic understanding of model performance (Park & Han, 2018; Reyna, Nsoesie, & Clifford, 2022; Hicks, 2022).

When implementing experimental replicates – which is typically recommended when working with small datasets common to microbiomics – performance of the model may vary due to random factors such as the exact split and the initialization of the model (Topçuoğlu et al., 2020). Thus when comparisons are made between multiple models, they should be validated through statistical testing. Similarly, if one is benchmarking the predictive value of microbiomic data against traditional risk factors, one must corroborate that a perceived difference is not simply due to random chance (e.g., sampling error). Notably, baseline and benchmark models provide important context for evaluating the predictive power of models and should be routinely incorporated into experimental design.

Post-hoc interpretability analyses, such as SHAP analysis, provide further valuable insights into how and why a model is making predictions (Lundberg & Lee, 2017). Meanwhile, error analyses provide insights into important demographic factors that affect model performance and can help identify potential failure modes to improve upon in future works (Nushi, 2021).

For further reading on various methods for model evaluation, we refer the reader to (Park & Han, 2018; Reyna, Nsoesie, & Clifford, 2022; Hicks, 2022). For additional discussion of post-hoc interpretability analyses we suggest (Reddy, 2022; Zednik & Boelsen, 2022). For discussion of error analysis we refer the reader to (Nushi, 2021).

Ensure that your work is reproducible

Reproducibility has been widely cited as a barrier to the advancement of science (Beam, Manrai & Ghassemi, 2020; Heil et al., 2021; Pineau et al., 2021). Here, we noted that key information about experimental methodology was often missing from the articles in our corpus, including complete descriptions of features used for ML, ML experimental design (e.g., nested vs repeated k-fold cross-validation), hyperparameter search sweeps and final settings, and seeds set for non-deterministic processes.

Sharing research code enhances transparency and reproducibility (Cadwallader, Mac Gabhann, Papin & Pitzer, 2022). Here, we found that 78% of papers refrained from publicly sharing their ML code. Even within the biomedical field, where 50.1% of studies do not share code (Sharma et al., 2024), this figure is concerning.

Techniques for enhancing reproducibility include using version-controlled repositories, sharing preprocessed datasets, documenting computational environments, applying open data and code sharing, performing random seed control, and providing clear documentation of code, amongst others. Figures demonstrating experimental design are also a powerful method for improving methodological transparency.

To combat inadequacies in reporting, which harms reproducibility, numerous reporting checklists have been developed for both study authors and reviewers. Biomedical study checklists include STORMS (Mirzayi et al., 2021), which is a microbiome-specific checklist, and clinical trial reporting checklists such as TRIPOD (Collins et al., 2015; Heus et al., 2019) and SPIRIT (Chan, Tetzlaff, Gøtzsche et al., 2013b, 2013a). In the ML domain, resources include the NeurIPS Paper Checklist (“NeurIPS 2021 Paper Checklist Guidelines” 2021) and the Papers With Code ML Code Completeness Checklist (Stojnic 2020). Some checklists span both ML and life sciences or healthcare, such as PROBAST (Wolff et al., 2019).

Furthermore, the adoption of workflow management systems can facilitate the creation of reproducible data analysis pipelines. These tools are designed to automate and streamline the execution of complex workflows, ensuring that every step is documented and consistently executed. Meanwhile various frameworks have been developed to help manage the machine learning lifecycle, providing functionalities for experiment tracking, version control, and model deployment.

For comprehensive discussion on reproducibility standards in ML, we refer the reader to (Pineau et al., 2021; Beam, Manrai & Ghassemi, 2020; Heil et al., 2021), in addition to the tools and reporting checklists described above.

Keep abreast of the latest research developments and explore emerging techniques

ML innovations are poised to tackle key challenges in the microbiome field. For example, small dataset sizes may be partially mitigated through the adoption of synthetic data augmentation techniques specifically for microbiomics data (Díez López, Montiel González, Vidaki, & Kayser, 2022; Mulenga, Kareem, Sabri, & Seera, 2021; Reiman & Dai, 2020; Sayyari, Kawas, & Mirarab, 2019; Gordon-Rodriguez, n.d.; Quinn, 2021) or novel embedding techniques for microbiome features that reduce the dimensionality of these datasets (Woloszynek, Zhao, Chen & Rosen, 2019). Fine-tuning of microbiome foundation models may one day become the norm, akin to work done on developing a foundation model for single-cell multi-omics data (Cui, Wang, Maan & Wang, 2023). Meanwhile, advancements in the integration and analysis of multi-omics data, including microbiomics data, will yield deeper insights into the dynamic interactions among various omics layers and allow for more accurate modeling of biological systems (Daliri, Ofosu, Chelliah, Lee, & Oh, 2021; Arkan & Muth, 2023).

Not all innovations will be unambiguously positive; some may expose vulnerabilities that could be exploited by malicious actors. For example, emerging evidence suggests training datasets can be reconstructed from the parameters of neural networks and other ML models (Balle, Cherubin, & Hayes, 2022; Wang & Kurz, 2022; Haim, n.d.), potentially compromising data governance and patient privacy.

Microbiomics practitioners can stay up to date on innovations in ML by attending ML conferences (e.g., the Conference on Neural Information Processing Systems (NeurIPS), the International Conference on Machine Learning (ICML), and the International Conference on Learning Representations (ICLR)), regularly reviewing new scientific publications (e.g., by setting up alerts to receive notifications on relevant research developments), and subscribing to newsletters from leading research institutions, amongst other methods.

Part 3: tutorial the application of ML to microbiomics data

To encourage the adoption of best practices in constructing ML

models for microbiome research, we developed an interactive tutorial that guides users through the development, evaluation, and interpretation of ML models for predicting whether an individual has schizophrenia based on the composition of their fecal microbiota. The goal of the tutorial is to evaluate: a) whether the microbiome has predictive value for the diagnosis of schizophrenia, and if so, b) which methodological ML parameters (i.e., learning algorithm, features) yield the highest predictive results, and c) which microbial taxa provide value for predicting whether an individual has schizophrenia. For this purpose, shotgun sequencing data was obtained for 171 fecal samples; 90 from medication-free individuals with diagnosed schizophrenia and 81 from individuals without diagnosed schizophrenia (Zhu et al., 2020). Repeated 10-fold CV was implemented to perform hyperparameter tuning and model training without data leakage. We applied SHAP to identify microbial taxa with the highest predictive value for schizophrenia and error analysis to gain insights into the distribution of errors across subsets of the study population. At each step, we provide detailed explanations of the rationale behind experimental design decisions, offer guidance on the application of best practices, and discuss the interpretation of the results. The tutorial is available at <https://github.com/quantori/tutorial-on-ML-for-microbiomics>.

Discussion

Microbiomics holds great potential for clinical applications, with ML set to transform our ability to gain meaningful insights from and harness the full utility of such datasets. Here, we employed a data-driven approach to characterize current practices in the application of ML to microbiomics data, identify where they diverge from best practices, and provide discussion on how to bridge this gap. To further support the application of ML best practices, we developed a tutorial tailored to the microbiomics community that provides an interactive demonstration of how to apply best practices when developing an ML model for microbiomics data.

Through our analysis of current practices in the application of ML to microbiomics, several issues emerged as field-wide challenges that will likely require concerted effort to overcome. The first is that the size of microbiome datasets tends to be relatively small for ML, likely resulting in high bias, high variance, and diminished model performance (Lakshmanan et al., 2020). In the present sample of the literature, the median dataset size was 161.5 samples, with 73% of studies having <1000 samples and over a third having <100 samples. In ML, the “rule of 10” approximates that dataset size is ideally on the order of ten times as many features as are present (Lakshmanan et al., 2020). If we use our tutorial on ML for microbiomics as an example, the final dataset which used microbial species as features consisted of 1082 features. This suggests that the ideal dataset size would be approximately 10,820 samples; 171 were available. It is essential that ML practitioners in the microbiomics field be aware of the deleterious effects of small dataset size and account for it during experimental design, model verification, and results interpretation (Vabalas, Gowen, Poliakoff & Casson, 2019; Ying, 2019). In the long term, data accumulation in repositories managed by large consortia will play a pivotal role in future efforts to source large datasets to power data-hungry ML algorithms. This data must be managed such that it is Findable, Accessible, Interoperable, and Reusable (FAIR) (Moreno-Indias et al., 2021). This responsibility extends beyond consortia; individual researchers must also contribute by submitting detailed and clear metadata alongside their data, and such behavior should be incentivized. Furthermore, the field must establish robust frameworks for developing and evaluating models that integrate data from multiple sources, where confounding variables are prevalent.

Another striking observation that resulted from our analysis was that important demographic data was often limited or missing in the literature reviewed here. This is problematic because characterizing the demographics of study participants provides context required to interpret study findings accurately, assess their relevance to broader populations,

and ensure that research is conducted fairly and ethically (Huang, Galal, Etemadi & Vaidyanathan, 2022; Perez-Downes et al., 2024; Starke, De Clercq & Elger, 2021; Vokinger, Feuerriegel & Kesselheim, 2021; Wiens et al., 2019). Here, we found that the most commonly reported demographic features were country of residence, age, and sex (76%, 72% and 66% of studies, respectively). Additional demographic attributes such as race/ethnicity, education level, and income were rarely, if ever, reported (11%, 2%, and 0% of studies, respectively). Such demographic factors are known to influence microbiome composition (Callahan et al., 2017; Chen et al., 2016; Kaplan et al., 2020; Ravel et al., 2011; Scepanovic et al., 2019), and thus have the potential to introduce bias into ML models. This could lead to models that perform well with one demographic but fail with others, posing a significant challenge when attempting to implement these models in clinical settings. For instance, we observed that studies involving individuals from the least studied continents (tied: Africa and Oceania) were 21 times less prevalent than those focusing on the most extensively studied continent (Asia). This disparity suggests that many of the models developed to date are likely to perform better for residents of Asia than for those of Africa and Oceania, given the distinct lifestyles, cultures, and genetic backgrounds of those populations. It is essential that ML for human health be developed and deployed in such a way that it performs equally well on diverse human populations (Huang et al., 2022; Perez-Downes et al., 2024; Starke et al., 2021; Vokinger et al., 2021; Wiens et al., 2019). Reporting study demographics is an important step towards assessing potential biases in models and identifying where additional research efforts are required, and it should be actively encouraged.

Finally, a concerning observation in our corpus of articles was an apparent widespread confusion over how to avoid test set omission and/or data leakage. Such a phenomenon has likely resulted in overestimates of model performance and the predictive value of the microbiome across the field. In fact, for 86% of papers, we were unable to confidently rule out test set omission and/or data leakage based on authors’ description of methodology. This finding is in close alignment with Quinn, (2021)’s estimate that up to 88% of studies applying ML to microbiomics datasets may contain test set omission and/or data leakage. We infer an urgent need to raise awareness within the microbiomics community about how to avoid and identify model verification failure in microbiomics studies. We refer the reader to Teschendorff, 2019; Teschendorff, 2019 and Whalen et al., 2022; Whalen, Schreiber, Noble & Pollard, 2022) for extensive discussion of what constitutes data leakage and how it can be avoided.

Acknowledging the limitations of this study is important for contextualizing its contributions and understanding the applicability of its findings in practice. First, we highlight that our review and analysis were based on a sample size of 100 peer-reviewed journal articles. While a larger sample size would have been advantageous, conducting a thorough review of each paper requires significant time and effort, particularly in instances where the methodology is complex, lengthy, and/or vaguely described, and we had to operate within the constraints of resources available to us. Despite the limited sample size, the trends identified in this study likely reflect fundamental issues within the field. We additionally recognize that the selection criteria used to identify papers for inclusion in any review will impact the study’s results. Here, we used PubMed to identify peer-reviewed articles with very few constraints, emulating the approach by which a novice reader may gather information. Alternative approaches would potentially yield different core findings, such as focusing on specific ML approaches or technologies, like deep learning, or on specific medical fields with unique characteristics and cultures. Finally, we recognize that our interpretations of the methodology employed by various studies were subjective in some cases, especially where manuscripts provided limited, vague, and/or conflicting descriptions. In order to mitigate the impact of this effect, each study surveyed for this analysis was reviewed by two ML experts. Interpretations of scientific methodology should not be subjective, and this issue highlights the importance of improving transparency and

reproducibility in science.

While this work focuses on the application of ML to microbiomics, we expect similar challenges are pervasive across other omics fields, such as genomics, transcriptomics, and metabolomics. For example, genomics research faces analogous issues with widespread data leakage (Teschendorff, 2019; Whalen, Schreiber, Noble, & Pollard, 2022) and underrepresentation of diverse populations in genomic datasets (Sirugo, Williams, & Tishkoff, 2019). We highlight two factors that may contribute to why such challenges may be shared across omics disciplines. First, the omics data can be costly, time consuming, and resource intensive to collect and generate. Therefore, many studies work with datasets that are relatively small for ML and/or incorporate data from public datasets (Li, Li, Xu, & Yang, 2022). We hypothesize that model verification failure often stems from confusion over how to apply ML best practices when implementing repeated cross-validation procedures in order to mitigate the deleterious effects of small dataset size. Furthermore, investigators may be unaware of how to implement cross-validation schemes that properly account for data being sourced from public databases covering multiple studies. Notably, sources of bias, such as racial disparities in data, stem from broader patterns in medical data collection practices, transcending any one omics discipline (Futoma, Simons, Panch, Doshi-Velez, & Celi, 2020; McCraden, Joshi, Mazwi, & Anderson, 2020; Obermeyer, Powers, Vogeli, & Mullainathan, 2019; Sirugo, Williams, & Tishkoff, 2019; Zou & Schiebinger, 2018). Second, the application of ML to omics data is highly interdisciplinary. Practitioners typically have a background in either bioinformatics or computer science, each possessing distinct educational foundations. For example, a traditional bioinformatician may not be well-versed in certain machine learning standards, such as the optimal quantity of data required for a specific ML task or the importance of hyperparameter tuning. Conversely, a machine learning expert may lack an understanding of the biological nuances associated with the features used in modeling, such as the advantages of log-ratio transformations in addressing the compositional nature of microbiomics data. Additionally, in the context of medical ML, neither group may be adequately familiar with identifying clinically useful problems or developing solutions that are not only scientifically sound but also ethically responsible and clinically applicable (Wiens et al., 2019). By drawing these parallels, we aim to emphasize that many of the issues and mitigation strategies highlighted here apply broadly across omics research and underscore the potential for cross-disciplinary solutions. In particular, there is significant potential for developing coursework, standards, tools, and frameworks that span multiple omics fields, thereby strengthening a broad spectrum of research.

The development of ML solutions in microbiomics and other areas of healthcare holds substantial potential for positive impact, yet it is also accompanied by numerous ethical concerns pertaining to patient rights, safety, and equity in care. Key considerations include data privacy and security, fairness, transparency and interpretability, and accountability (Char, Abramoff, & Feudtner, 2020; Nassar & Kamal, 2021; Zhang & Zhang, 2023). ML is inherently dependent on large datasets, which implies that the methodology used in dataset construction entails significant ethical considerations. Obtaining informed consent from individuals for the use of their data in the development of ML models can be complex, particularly when working with data sourced from large databases that were originally compiled for different purposes (Iserson, 2024; Larson, Magnus, Lungren, Shah, & Langlotz, 2020; Thapa & Camtepe, 2021). Protection of said data is also a major concern, with risks including data breaches, unauthorized access, and misuse of patient data (Thapa & Camtepe, 2021). Furthermore, biases in the composition of datasets can adversely affect the model's predictive performance across different demographic groups, potentially exacerbating healthcare disparities if these models are not deployed and made accessible in an equitable manner (McCradden, Joshi, Mazwi, & Anderson, 2020; Obermeyer, Powers, Vogeli, & Mullainathan, 2019; Zou & Schiebinger, 2018). There are also important considerations around how

to ethically apply ML to patients in the clinical setting, particularly concerning transparency, interpretability, and accountability. Many models operate as black boxes, meaning clinicians are unaware of their underlying decision-making process. This lack of visibility poses challenges when applying model recommendations, as clinicians may struggle to identify potential errors or biases within the predictions (Amann, Blasimme, Vayena, Frey, & Madai, 2020; Cuttillo, 2020). Consequently, questions also arise over accountability. When medical errors arise due to faulty ML predictions, it may be unclear whether responsibility lies with the model developers, the clinician, or the healthcare institution (Naik, 2022). As a society, we are still in the process of optimizing approaches to address these ethical questions. It is important for research communities to remain informed and actively engaged in the development of ethical frameworks and guidelines, while regulatory agencies establish comprehensive regulations governing the use of ML in healthcare.

Awareness of widespread deviations from ML best practices within the microbiomics field can be used to improve the quality and reliability of research, in some cases through relatively simple interventions. As ML practitioners, reviewers, and/or regulators, it is our responsibility to optimize the use of scientific resources and ensure that developed ML models are effective, reliable, trustworthy, and fair. Applying best practices to scientific research is essential for expediting the advancement of healthcare and improving patient lives.

Credit author statement

Natasha Dudek: Conceptualization, Methodology, Software, Formal Analysis, Writing - Original, Draft, Writing - Review & Editing, Visualization. **Mariami Chakhvadze:** Methodology, Software, Formal Analysis, Writing - Review & Editing. **Saba Kobakhidze:** Formal Analysis, Writing -Review & Editing. **Omar Kantidze:** Conceptualization, Methodology, Writing - Review & Editing, Supervision. **Yuriy Gankin:** Conceptualization, Writing - Review & Editing, Supervision,Resources.

Data availability statement

Data used in the interactive tutorial was sourced from (Zhu et al., 2020) and is available via the European Nucleotide Archive (ENA) database, accession number ERP111403.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.mlwa.2024.100607.

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